

Hands-On MIS Projects

Chapter 1 - Information Systems in Global Business Today

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1 Introduction

This report analyzes management information system problems and demonstrates how information systems improve decision making using transactional sales data.

2 Management Decision Problem 1-7: Snyder's of Hanover

2.1 Problem Summary

Snyder's financial department depended heavily on spreadsheets and manual processes for collecting and reporting financial information. Every month, the financial analyst collected spreadsheets from more than 50 departments worldwide, consolidated the data, and created the company's profit-and-loss statement manually. This process was slow, difficult to maintain, and vulnerable to errors.

2.2 Impact on Business Performance

The spreadsheet-based system negatively affected business performance.

- The financial analyst spent significant time collecting and consolidating spreadsheets instead of performing financial analysis.
- Manual data entry increased the possibility of errors and inconsistent reports.
- Financial information was delayed because reports required manual preparation.
- The company faced higher operational costs because employees performed repetitive administrative tasks.
- The system was difficult to scale as the company expanded globally.

2.3 Impact on Management Decision Making

The existing system limited management decision making because managers did not have fast access to accurate and updated financial information.

Delayed reporting made it harder to identify financial problems, compare department performance, and respond quickly to changes in business conditions. Incorrect or outdated information could result in poor decisions regarding budgeting, production, and resource allocation.

2.4 Recommended Information System Solution

Snyder's should implement an integrated financial information system based on a centralized database. The system should allow departments to enter financial data directly into a shared platform and automatically generate standardized reports.

The system should include automated reporting, security controls, and management dashboards. An Enterprise Resource Planning (ERP) system could integrate financial data with other business functions such as sales, inventory, and production.

This solution would reduce manual work, improve reporting accuracy, provide faster access to information, and support better management decisions.

3 Management Decision Problem 1-8: Dollar General

3.1 Problem Summary

Dollar General operates discount stores where minimizing costs is essential. However, the company does not have an automated inventory tracking system at each store. Managers cannot accurately verify incoming deliveries, which has resulted in inventory errors and increasing merchandise losses.

The company needs to determine whether an information system investment can improve inventory accuracy, reduce losses, and support better operational decisions.

3.2 Decisions Before Investing in an Information System

Before investing in an inventory management system, Dollar General should evaluate several important factors.

- **Identify business goals:** The company should define the problems it wants to solve, such as reducing inventory losses, improving inventory accuracy, and ensuring product availability.
- **Determine required system features:** The company should decide whether it needs barcode scanning, real-time inventory tracking, automatic stock updates, inventory alerts, and reporting tools.
- **Evaluate costs and benefits:** Management should compare implementation costs, including hardware, software, and training, against expected benefits such as lower losses and improved operational efficiency.
- **Plan implementation:** The company must decide how the system will be introduced, how employees will be trained, and how existing inventory data will be transferred.
- **Consider security requirements:** The company should establish rules for protecting inventory data and controlling access to the system.

Dollar General should implement an automated inventory management system using barcode scanning and a centralized database. This system would provide accurate inventory information, reduce merchandise losses, improve replenishment decisions, and increase overall operational efficiency.

4 Database Analysis Project 1-9

4.1 Dataset Description

The project uses a transactional marketplace database stored in SQLite format. The database contains raw sales transaction data that can be used to analyze product demand, regional sales performance, seller performance, and monthly sales trends.

The database contains five main tables:

- **orders**: contains order identification numbers, order dates, and customer contact information.
- **order_items**: contains the products sold in each order, seller identifiers, product identifiers, and item prices.
- **products**: contains product identifiers, product categories, and product weight information.
- **sellers**: contains seller identifiers and seller state information.
- **order_reviews**: contains customer review scores and review comments.

The main analysis used the **orders**, **order_items**, **products**, and **sellers** tables. The **order_reviews** table was not required for the three main sales trend questions, but it could be used in future analysis to study customer satisfaction.

The dataset included 96,478 orders, 110,197 order item records, 32,216 products, 2,970 sellers, and 39,462 reviews. The order dates ranged from September 15, 2016 to August 29, 2018.

Table	Number of Records
orders	96,478
order_items	110,197
products	32,216
sellers	2,970
order_reviews	39,462

Table 1: Database Tables and Record Counts

4.2 Data Preparation

The original dataset was provided as a SQLite database file. Because the project requires database querying and reporting, SQLite was used to inspect the database structure and run SQL queries. The database was also exported to CSV files so the tables could be opened in spreadsheet or database software if needed.

A Python script was created to export each SQLite table into a separate CSV file. The exported files were:

- `orders.csv`
- `order_items.csv`
- `products.csv`
- `sellers.csv`
- `order_reviews.csv`

The main sales analysis was performed by joining the order, product, and seller tables into a single SQL view named `sales_analysis_base`. This view created a structure similar to the textbook sales database.

Textbook Field	Dataset Field Used
Store identification number	<code>seller_id</code>
Sales region	<code>seller_state</code>
Item number	<code>product_id</code>
Item description	<code>product_category_name</code>
Unit price	<code>price</code>
Units sold	Count of records in <code>order_items</code>
Sales period	Month extracted from <code>timestamp</code>

Table 2: Mapping Textbook Sales Fields to Dataset Fields

A missing data check was also performed on the joined sales view. The analysis found no missing product categories, seller states, or prices in the final analysis table.

4.3 Sales Analysis Queries

SQL queries were used to answer the three management questions in the assignment. The queries joined the transactional tables and summarized the data by product category, seller region, seller, and month.

The first query group identified which products should be restocked by ranking product categories and individual products by units sold and total revenue. Since the dataset did not include inventory quantity, high sales volume was used as a proxy for high customer demand.

The second query group identified stores and regions that may benefit from promotional campaigns. In this dataset, `seller_id` represented the store or seller, and `seller_state` represented the sales region. Regions with low revenue and low sales volume were treated as candidates for additional marketing support.

The third query group analyzed monthly sales trends. Since the dataset did not include a direct discount field, monthly revenue and units sold were used to identify stronger and weaker sales periods. Stronger months were considered better candidates for full-price selling, while weaker months were considered better candidates for discounts or promotional campaigns.

4.4 Restocking Analysis

The first management question asks which products should be restocked. The dataset does not include current inventory levels or stock quantities. Therefore, units sold were used as a proxy for customer demand. Product categories with the highest sales volume were treated as the strongest candidates for restocking.

The analysis showed that the highest-demand product categories were `cama_mesa_banho`, `beleza_saude`, `esporte_lazer`, `moveis_decoracao`, and `informatica_acessorios`.

Product Category	Units Sold	Total Revenue
<code>cama_mesa_banho</code>	10,953	1,023,434.76
<code>beleza_saude</code>	9,465	1,233,131.72
<code>esporte_lazer</code>	8,431	954,852.55
<code>moveis_decoracao</code>	8,160	711,927.69
<code>informatica_acessorios</code>	7,644	888,724.61

Table 3: Top Product Categories by Units Sold

Based on this analysis, managers should prioritize restocking these high-demand categories. The category `beleza_saude` is especially important because it has both high sales volume and the highest total revenue among the top categories. This suggests that keeping this category available could have a significant impact on sales performance.

4.5 Promotional Campaign Analysis

The second management question asks which stores and sales regions would benefit from a promotional campaign and additional marketing. In this dataset, `seller_id` was used as the store identifier, and `seller_state` was used as the sales region.

The analysis compared sales performance by region using units sold, number of sellers, total revenue, and average price. Regions with low total revenue were identified as possible candidates for promotional campaigns or additional marketing support.

Region	Units Sold	Number of Sellers	Total Revenue
AM	3	1	1,177.00
PA	8	1	1,238.00
SE	10	2	1,606.20
PI	11	1	2,383.00
RO	14	2	4,762.20
MS	50	5	8,551.69
RN	56	5	9,992.60
PB	37	6	16,395.50
MT	144	4	17,007.82
CE	90	12	19,517.84

Table 4: Lowest-Performing Sales Regions by Revenue

The regions AM, PA, SE, PI, and RO had the lowest total revenue. These regions may benefit from additional marketing campaigns because their sales activity is much lower than the strongest regions.

However, the results should be interpreted carefully. Some of these regions have only one or two sellers, which suggests that low revenue may be caused by weak market presence rather than only weak customer demand. Therefore, management should consider both promotional campaigns and seller expansion strategies in these regions.

For comparison, the strongest region was SP, with 78,604 units sold and 8,509,511.46 in total revenue. This shows a large performance gap between SP and the lower-performing regions.

4.6 Pricing and Discount Timing Analysis

The third management question asks when products should be offered at full price and when discounts should be used. The dataset does not contain a direct discount field, so monthly sales volume and total revenue were used to identify stronger and weaker sales periods.

High-demand months are better candidates for full-price selling because customers are already purchasing more products. Low-demand months are better candidates for discounts, promotions, or additional marketing because sales activity is weaker.

Month	Units Sold	Total Revenue
May	11,814	1,466,882.94
August	11,939	1,393,276.34
July	11,379	1,349,557.98
April	10,396	1,314,203.77
March	10,914	1,312,555.10

Table 5: Strongest Months by Total Revenue

Based on this result, managers should consider selling products closer to full price during March, April, May, July, and August. These months had the strongest sales performance and may not require aggressive discounting.

Month	Units Sold	Total Revenue
September	4,740	607,534.64
October	5,527	688,572.76
December	6,188	726,044.09
November	8,475	987,765.37
January	8,950	1,036,443.36

Table 6: Weakest Months by Total Revenue

Discounts and promotional campaigns should be considered during September, October, December, November, and January because these months had weaker sales performance. Promotions during these periods may help increase demand and improve revenue.

The results should be interpreted carefully because the dataset does not cover all calendar months equally. The data begins in September 2016 and ends in August 2018, so some periods may be incomplete. Even with this limitation, the monthly trend provides useful guidance for pricing and promotional planning.

5 Job Search Project 1-10

6 Conclusion